

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414486
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Gomez; Manuel M.

360 degree landscaping tool

Abstract

A landscaping tool comprising a disc rotatably attached to an attachment end of an arm. The disc comprises a sharp edge, allowing for cutting, breaking, severing, etc. of various landscape features during lateral and rotational movement of the disc. The arm further comprises a handle and an angled section. The handle allows a user to grasp the arm and swing or otherwise control the arm and disc. The angled section of the arm creates an ideal angle between the disc and arm, allowing for most effective use of the invention for cutting and other landscaping purposes.

Inventors:	Gomez; Manuel M. (Twain Harte, CA)
Applicant:	Gomez; Manuel M. (Twain Harte, CA)
Family ID:	1000008819340
Appl. No.:	17/825461
Filed:	May 26, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220377957 A1	Dec. 01, 2022

Related U.S. Application Data

us-provisional-application US 63193955 20210527

Publication Classification

Int. Cl.: A01B1/08 (20060101); A01B1/22 (20060101); A01G3/06 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: A01B (1/08); A01B (1/22); A01D (34/84); A01G (3/06); B25G (1/04)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
620909	12/1898	Gerlach	172/182	A01G 3/06
910111	12/1908	Carter	172/16	A01G 3/06
1763339	12/1929	Beck	172/15	A01G 3/06
2506366	12/1949	Korn	172/15	A01G 3/06
2634667	12/1952	Woolwine	172/186	A01G 3/06
2890525	12/1958	Altendorf	172/15	A01G 3/06
3139940	12/1963	Randell	172/15	A01G 3/06
5951078	12/1998	Whitehead	294/54.5	E01H 5/02
6516598	12/2002	Notaras	56/16.7	A01D 34/001

Primary Examiner: Sebesta; Christopher J

Assistant Examiner: Lusk; Audrey L

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates generally to a landscaping tool to cultivate the surrounding land. More specifically, the present invention is a device that utilizes a concave disc **1** to manipulate the ground easily and efficiently.

BACKGROUND OF THE INVENTION

(2) For many years landscapers and gardeners have utilized and stuck with the same tools for manipulating the ground in the desired way. Over countless years many of these tools have not grown and developed much along with the development of technology. This leaves individuals stuck with using tools that are limited in only performing about on to three functions. Using a hoe tool for instance can only be used to remove weeds by shaving them off and cannot be utilized to be an edger, to scoop the dirt in the ground, to create planting furrows, and to cultivate the ground for planting. The other issue with many regular garden and landscaping tools is that many of them are limited to one size. This causes a problem for people who are of different heights and strengths attempting to use the same tool. A tool that is not properly size for an individual can result in less efficient use of the tool or even in potential injury by misusing the tool. Many of these individuals require a tool that allows for versatility in the way it can be used and versatility in the usable length.

(3) An objective of the present invention is to provide users with a landscaping tool, to help cultivate land in various ways efficiently and easily. The present invention intends to provide users with a device that rotates a concave disc to preform various landscaping needs. In order to accomplish that, a preferred embodiment of the present invention comprises a disc, an arm, and a

handle. Further, the handle utilizes a telescopic section to ensure the present invention can be sized for any individual. The present invention will allow the user to improve with the tool and use it in various situations and in various ways. By having alternative embodiments of the present invention with a sturdy 6 ft pole as a one of the telescopic sections, the present invention can be used in multiple different ways. Thus, the present invention is a landscaping tool that can be adjusted to various lengths to ensure the present invention can cultivate land in an efficient and convenient way.

SUMMARY OF THE INVENTION

(4) The present invention is a landscaping and gardening tool to help with various jobs that would normally require a variety of different tools. The present invention seeks to provide users with a device that adjusts in length and utilizes various sized and shaped discs in order to efficiently manipulate the desired land being worked upon. In order to accomplish this the present invention comprises a disc that can rotate 360 degrees to easily dig into the ground, because the 360 degree blade is constantly moving, cutting and working with all edges of the disc. Further, the arm holds the disc and allows it to rotate without the rotation causing the nylon nut to fall off or lock up the disc. Additionally, the handle utilizes a telescopic section to ensure the present invention can be used with people of various heights. Thus, the present invention is a landscaping tool that can be adjusted to various lengths to ensure the present invention can cultivate land in an efficient and convenient way.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top front perspective view of the present invention in accordance with at least one embodiment.
- (2) FIG. 2 is a side elevation view of the present invention in accordance with at least one embodiment.
- (3) FIG. 3 is a rear elevation view of the present invention in accordance with at least one embodiment.
- (4) FIG. 4 is a front elevation view of the present invention in accordance with at least one embodiment.
- (5) FIG. 5 is a bottom plan view of the present invention in accordance with at least one embodiment.
- (6) FIG. 6 is a top plan view of the present invention in accordance with at least one embodiment.
- (7) FIG. 7 is a side view of the angled section of the present invention in accordance with at least one embodiment.
- (8) FIG. 8 is a side view of the disc of the present invention in accordance with at least one embodiment.
- (9) FIG. 9 is a top view of the disc of the present invention in accordance with at least one embodiment.

DETAIL DESCRIPTIONS OF THE INVENTION

- (10) All illustrations of the drawings are for the purpose of describing selected versions of the present invention and are not intended to limit the scope of the present invention.
- (11) As shown in FIG. 1, the present invention is a landscaping tool. An objective of the present invention is to provide users with a tool that can utilize various discs and lengths to ensure the present invention can be used in various landscaping and gardening scenarios. To accomplish this, the present invention comprises a disc **1**, an arm **2**, and a handle **21**. Many of these components allow for the present invention to utilize the disc **1** by rotating the bladed edge **11** of the disc 360 degrees to cut into the surrounding ground. The disc **1** is placed on the top of the arm **2** via the disc

hole **12**. Located on the arm **2**, the handle **21** is securely fixed. The nylon nut **13** allows the disc **1** to rotate freely around the arm **2** without loosening or tightening the nylon nut **13** and without falling off the arm **2**. Thus, the present invention is a landscaping tool that can be adjusted to various lengths to ensure the present invention can cultivate land in an efficient and convenient way.

(12) The present invention can cut through ground during landscaping and gardening projects with the disc **1** component. The disc **1** is made of a top-grade steel material with a circular concave shape as seen in FIG. **1**, **6**, **8-9**. The concave angle **14** of the disc **1** is between 55 and 85 degrees as shown in FIG. **8**. The disc **1** is designed with a concave shape where the edge **11** around the disc **1** is pointed into a blade shape that allows the disc **1** to easily cut through ground, roots, etc. In its preferred embodiment the disc **1** comprises a washer, a nylon nut **13**, and a disc hole **12**. Referring to FIG. **9**, the disc **1** can come in various sizes with the diameter **15** of the disc **1** ranging from two to six inches to be used effectively depending on the landscaping project. The washer may be located at the concentric center of the disc **1** as shown in FIG. **6**. The washer is made of a metal material and has a wide ring shape with a thin thickness, designed to endure most of the grinding between materials as the disc **1** rotates around the arm **2** to ensure the longevity of the tool. The nylon nut **13** attaches at the attachment end **23** of the arm **2** and holds the disc **1** in place along the arm **2**, only allowing for rotational movement. The nylon nut **13** also ensures that as the disc **1** rotates it will not loosen or tighten the nylon nut **13** which would either disable rotational movement or potentially allow the disc **1** to fall off. To ensure that the disc **1** can be placed on the arm **2** properly, the disc hole **12** is designed wherein the center of the disc **1** has a diameter similar to the attachment end **23** of the arm **2**. This disc hole **12** is designed to allow the disc **1** to be snugly placed on the arm **2** while allowing for rotational movement around it. It should be further noted that, the disc **1** can be created in many various shapes and sizes and the washer can be positioned in various ways while still staying within the scope of the present invention.

(13) For example, while the disc **1** generally has a circular shape, a disk **1** of the present invention could be square, triangular, or any other shape. Additionally, edge **11** of the disk **1** may be serrated, jagged, or shaped in another manner different from the uniform circular shape of the preferred embodiment.

(14) The arm **2** holds the disc **1** via the attachment end **23** of the arm **2**. The arm **2** is made of a steel material with an angled section **22**. The angle **28** of the angled section **22** is preferably 135 degrees as seen in FIG. **4**. This angle **28** of the angled section **22** is not limited to just 135 degrees and could be larger or smaller without straying from the scope of the present invention, with the most useful range being 135-150 degrees. This angle **28** allows the disc **1** to be rotated in a more efficient manner along the ground. In its preferred embodiment the arm **2** comprises an attachment end **23**, a disc holder **24**, a hollow section and a screw hole. The attachment end **23** is designed as a spiral cutout that has a mating thread to match the thread of the nylon nut **13**. This allows the nylon nut **13** to be properly secured to the arm **2**. Below the attachment end **23** is the disc holder **24**. The disc holder **24** is located before the angled section **22** of the arm **2** as seen in FIG. **5**. This design allows for the disc **1** to be secured on one end with the nylon nut **13** and limited from falling down the arm **2** by the disc holder **24** which has a larger diameter than the disc hole **12**. By sandwiching the disc **1** between the nylon nut **13** and the disc holder **24**, the disc **1** is secured to the arm **2** and resistant to all movement other than rotational movement about the arm **2**. At the bottom half of the arm **2** is the hollow section that allows for the arm **2** to be connected with the handle **21**. In order to ensure the handle **21** stays secured onto the arm **2** a screw hole may be positioned on the rear side.

(15) The handle **21** connects with the arm **2** via the portion of the arm **2** below the angled section **22**. The handle **21** is made of a durable material with a long cylindrical shaft design. In its preferred embodiment the handle **21** comprises at least one telescopic section **25**, including a plurality of holes **26**, and at least one spring mechanism **27**. The telescopic section **25** is designed as a cylindrical shaft with a diameter slightly smaller than that of the handle **21** allowing for it to slide

in and out of the handle **21** to increase or decrease the overall length of the present invention. This design allows for the handle **21** to extend to a greater length to ensure the present invention can be comfortably used by people of various heights with the same efficiency and ease. The telescopic section **25** may be at either end of the handle **21**. In one embodiment, there may be a telescopic section **25** attached to the arm **2** near the angled section **22** as well as opposite the angled section **22**. In an alternative embodiment the present invention can be designed with an ergonomic handle **21** that can be gripped and held easily by the user. This will allow the present invention to be used for close up precision work where the individual needs to get on their hands and knees to correctly work on the surrounding ground. The plurality of holes **26** is positioned on the rear side of the telescopic section **25** or the handle **21**. This design allows for the telescopic section **25** to be extended and retracted to predetermined lengths. The plurality of holes **26** combined with the at least one spring mechanism **27** ensures that the handle **21** can be extended to a certain length and then locked into the desired length. Once the present invention has been used at the desired length and needs to be changed to a different length, one of the at least one spring mechanism **27** can be pushed, unlocking the telescopic section **25** from one of the plurality of holes **26** and extended or retracted to another of the plurality of holes **26** and locked into place again. With all the various components working in tandem with each other it can be seen that the present invention is a landscaping tool that can be adjusted to various lengths to ensure the present invention can cultivate land in an efficient and convenient way.

(16) Although the invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention.

Claims

1. A landscaping tool, comprising: a disc, the disc comprising: an edge, the edge of the disc being pointed into a blade shape and being rotatable 360 degrees to cut into surrounding ground; an arm; the arm further comprising a handle, an angled section, and an attachment end; the handle being oppositely positioned on the arm from the attachment end; the angled section positioned adjacent to the attachment end, the angled section being 135 degrees relative to the arm thereby allowing the disc to be rotated efficiently along the ground to cut into surrounding ground; the disc being freely rotatably attached to the attachment end of the arm; a disc holder attached to the arm between the angled section and the attachment end; a nylon nut attached to the attachment end of the arm; the disc further comprising a disc hole in a center of the disc; the disc hole positioned on the attachment end of the arm between the disc holder and the nylon nut, the nylon nut holding the disc in place with a washer along the arm and only allowing rotational movement as the disc rotates 360 degrees to cut into surrounding ground and ensuring that the disc does not loosen or tighten that would disable rotational movement of the disc; and the disc hole being smaller in diameter than the disc holder allowing secure attachment of the disc to the arm allowing rotational movement of the disc about the arm as the disc rotates 360 degrees to cut into surrounding ground and preventing non-rotational movement.
2. The landscaping tool of claim 1, wherein the disc has a circular concave shape comprising a concave angle of the disc.
3. The landscaping tool of claim 2, wherein the concave angle of the disc is between 55 and 85 degrees.
4. The landscaping tool of claim 1, further comprising: at least one telescopic section being adjustably slidably attached to the handle and configured to extend and retract relative to the handle.
5. The landscaping tool of claim 4, further comprising: the handle further comprising a plurality of holes; the at least one telescopic section further comprising a spring mechanism; and the spring

mechanism being insertably compatible with the plurality of holes.

6. The landscaping tool of claim 1, wherein the diameter of the disc is between 2 and 6 inches.

7. A landscaping tool, comprising: a disc, the disc comprising: an edge, the edge of the disc being pointed into a blade shape and being rotatable 360 degrees to cut into surrounding ground; an arm; the arm further comprising a handle, an angled section, an attachment end, and a disc holder; the handle being oppositely positioned on the arm from the attachment end; a disc holder attached to the arm between the angled section and the attachment end; the angled section and the disc holder each positioned adjacent to the attachment end, the angled section being 135 degrees relative to the arm thereby allowing the disc to be rotated efficiently along the ground to cut into surrounding ground; the disc being freely rotatably attached to the attachment end of the arm; a nylon nut attached to the attachment end of the arm; the disc further comprising a disc hole in a center of the disc; the disc hole positioned on the attachment end of the arm between the disc holder and the nylon nut, the nylon nut holding the disc in place with a washer along the arm and only allowing rotational movement as the disc rotates 360 degrees to cut into surrounding ground and ensuring that the disc does not loosen or tighten that would disable rotational movement of the disc; and the disc hole being smaller in diameter than the disc holder allowing secure attachment of the disc to the arm allowing rotational movement of the disc about the arm as the disc rotates 360 degrees to cut into surrounding ground and preventing non-rotational movement.

8. The landscaping tool of claim 7, wherein: the disc has a circular concave shape; and the diameter of the disc is between 2 and 6 inches.

9. The landscaping tool of claim 7, wherein: the concave angle of the disc is between 55 and 85 degrees.

10. The landscaping tool of claim 7, further comprising: the handle further comprising at least one telescopic section adjustably slidably attached to the handle and configured to extend and retract relative to the handle; the handle further comprising an ergonomic grip; the handle further comprising a plurality of holes; the at least one telescopic section further comprising a spring mechanism; and the spring mechanism being insertably compatible with the plurality of holes.

11. A landscaping tool, comprising: a disc, the disc comprising: an edge, the edge of the disc being pointed into a blade shape and being rotatable 360 degrees to cut into surrounding ground; an arm; the arm further comprising a handle, an angled section, an attachment end, and a disc holder, the angled section being 135 degrees relative to the arm thereby allowing the disc to be rotated efficiently along the ground to cut into surrounding ground; a disc holder attached to the arm between the angled section and the attachment end; a nylon nut attached to the attachment end of the arm; the disc further comprising a disc hole in a center of the disc; the disc hole positioned on the attachment end of the arm between the disc holder and the nylon nut, the nylon nut holding the disc in place with a washer along the arm and only allowing rotational movement as the disc rotates 360 degrees to cut into surrounding ground and ensuring that the disc does not loosen or tighten that would disable rotational movement of the disc; and the disc hole being smaller in diameter than the disc holder allowing secure attachment of the disc to the arm allowing rotational movement of the disc about the arm as the disc rotates 360 degrees to cut into surrounding ground and preventing non-rotational movement the disc has a circular concave shape; and the diameter of the disc is between 2 and 6 inches.

12. The landscaping tool of claim 11, further comprising: the handle further comprising at least one telescopic section adjustably slidably attached to the handle and configured to extend and retract relative to the handle; the handle further comprising an ergonomic grip; the handle further comprising a plurality of holes; the at least one telescopic section further comprising a spring mechanism; and the spring mechanism being insertably compatible with the plurality of holes.

13. The landscaping tool of claim 12, wherein: the concave angle of the disc is between 55 and 85 degrees.
